Quantum Technologies for Autonomous Vehicles

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Table of Contents

I. Introduction/ Background 4

II. Literature Review 7

III. Problem Definition 14

IV. Project Plan 17

V. Conclusion 30

VI. References 32

Abstract— **With plenty of research being conducted to form a deeper understanding of quantum science, more specifically quantum computing, there have been few studies and integrating quantum technologies into more traditional robotics and autonomous systems. The overall goal of this project is to experiment with the quantum mechanical phenomenon of quantum entanglement for robotics and autonomy. Known as Quantum Key Distribution (QKD), quantum cryptography is the application of quantum entanglement in which the state of two or more particles cannot be differentiated from one another until it is measured, thus any attempt of interception will lead to the eavesdropper to be detected. By sending a stream of entangled polarized photons through a physical channel or free space, two parties (Alice and Bob), can securely transmit information by generating a secret key. Encrypted data is then transmitted which will contain the information used by Alice and Bob robots for classical controls and autonomy by using the key to decrypt the commands. The focus of this paper is to discuss object tracking, more specifically using a combination of fine/coarse tracking methods.**

Keywords—*Quantum Cryptography, Quantum Key Distribution, Quantum Multibody Dynamics, Quantum Entanglement, Quantum Key Distribution (QKD), Coarse Tracking, Fine Tracking*

I. Introduction / Background

Quantum cryptography is a promising new technology with the potential to change the future of communication. Although it is still in its early stages of development, quantum cryptography has the potential to revolutionize the way sensitive information is shared. With a project as large as this, many of the key components are broken up among students so that small groups can effectively work on their task. The focus of this paper is on the emphasis of object tracking through the usage of fine and coarse alignment to help maintain system integrity for quantum cryptographic communication. To understand why this portion of the project is important, a brief explanation on what cryptography is and how it is enhanced by quantum entanglement must first be explored.

Modern cryptography refers to a system in which a message is enciphered using a privately disclosed key. This key allows the user to send a secured message to the receiver that has their private deciphering key. That message cannot be compromised through traditional forms of eavesdropping and is computationally secure as the methods of deciphering are too complex to be economically viable [2]. The system utilizes 2 ground vehicles Alice the transmitter, and Bob the receiver. Quantum cryptography can be summarized using the following analogy; Alice places an object in a container and sends it to Bob where he locks the container using his own lock and sends it back to Alice. Alice unlocks Bob's lock and sends the container back to him where he can unlock his own lock to get the object [19]. However, with recent advances in computational technologies, it may soon be economically viable to crack encrypted messages sent through public channels and try a multitude of randomly generated passwords and keys [3].

As a result of this, principles of quantum mechanics are utilized to generate a secure key for the encryption and decryption of messages. Quantum Key Distribution (QKD) uses the properties of quantum entanglement to generate randomly polarized photons in either a horizontal or vertical

state known as Qubits. What makes them much more secure than classical bits (1s and 0s) is that the qubits can be in two states at the same time. The state of the photon is unknown until it is measured after passing through a polarizer on Bob with a predetermined basis (meaning if it is horizontal/vertical or diagonal polarizer), making it much easier to identify an eavesdropper as it will disturb the qubits which will be detected by both sender and receiver.

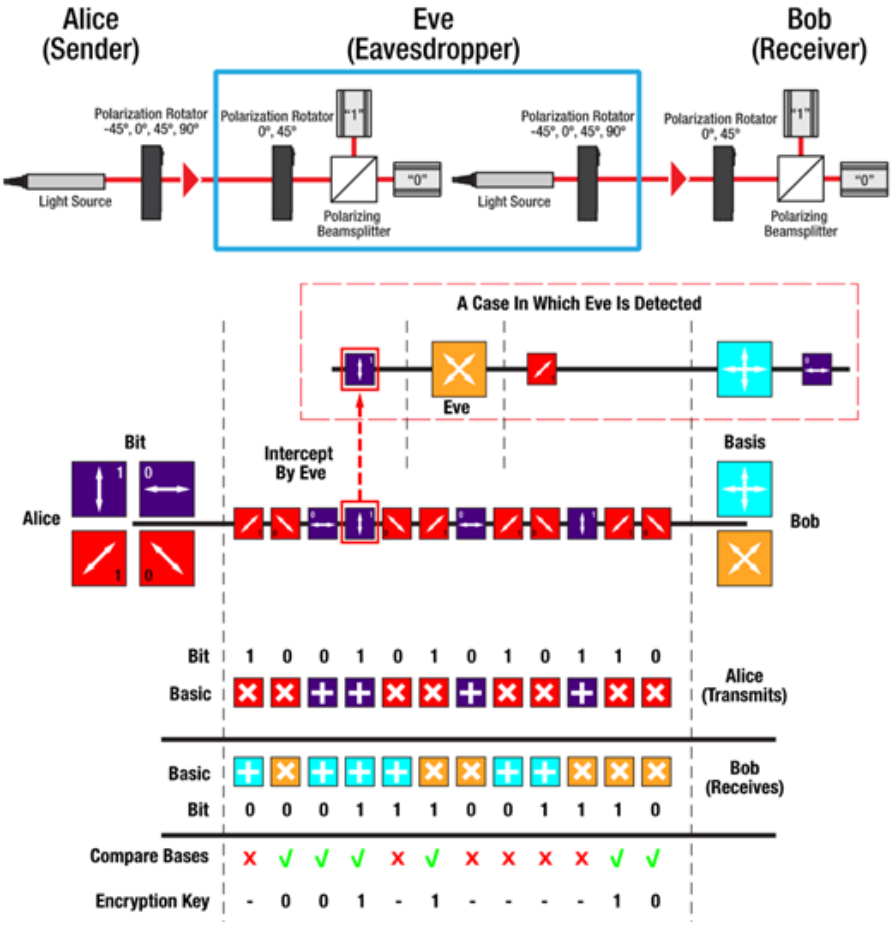


Fig. 1 Thorlab, Quantum Cryptography Demonstration Kit: User Manual.

This is due to the no cloning theorem where replication of an arbitrary unknown quantum state is impossible. [22] This is to say, it is impossible to derive a measurement from a simple observation of a quantum state. [23] The quantum state must be processed through secondary systems to be measured, an interaction with the qubits will not result in a meaningful result.

Continuing on, the polarization of the photon as a binary bit to generate the key which will serve as the command for controlling the robot [1][7]. Each drone will compare the measurements on parts of the qubits. Using key shifting, Alice and Bob will decide which parts of the qubits to use. Due to decoherence, the loss of the superposition of the photons is possible and irreversible, so the system must check for an error rate. If the error rate is too high, the system will know that there is an eavesdropper or that the signal has degraded and will generate a new key until the error rate is satisfied.

This method of cryptography also then serves as a secure layer of protection to mission-critical devices such as UAVs for law enforcement, defense, and disaster recovery from being affected by adversaries who will try to disrupt communications between the classical systems that use radio via jamming, spoofing, etc [10]. To help create a line of communication between the two autonomous robots a third drone will be used as a node for quantum key distribution. This approach will allow a multitude of cryptographically controlled vehicles to communicate with one another easily as the drone can help reduce signal degradation due to distance and weather by being at a higher altitude [4][6].

II. Literature Review

Farbod Khoshnoud, Maziar Ghazinejad, Automated quantum entanglement and cryptography for networks of robotic systems, IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA), IDETC-CIE 2021, Virtual Conference: August 17 – 20, DETC2021-71653, 2021.

"Procedure for automating the photon quantum experiments for mobile robotic applications… Quantum cryptography, single photons are sent from Alice robot to Bob robot, where Alice has the course of single photon and Bob has a polarizing beamsplitter and two detectors and that can detect the polarization of photons as vertical and horizontal. Bob then can convert the polarizations to a digital signals as zeros and ones and use them as communication information for control purposes through a classical channel. Motorized optics equipment can automatically align the source of photons to detectors on the mobile robots. The automated alignment procedure is one of the key enabling technologies in integrating quantum capabilities with control of mobile robotic systems."

Farbod Khoshnoud, Lucas Lamata (Sevilla), Clarence W. De Silva (UBC), Marco B. Quadrelli (JPL), Quantum Teleportation for Control of Dynamical Systems and Autonomy, Journal of Mechatronic Systems and Control, Volume 49, Issue 3, pp. 124-131, 2021.

"Review of how Quantum Entanglement and Quantum Cryptography can be integrated

into macroscopic mechanical systems for controls and autonomy applications is presented, as well as how quantum teleportation concepts may be applied to the classical domain. In quantum teleportation, an entangled pair of photons which are correlated in their polarizations are generated and sent to two autonomous platforms, which we call the Alice Robot and the Bob Robot. Alice has been given a quantum system, i.e. a photon, prepared in an unknown state, in addition to receiving an entangled photon. Alice measures the state of her entangled photon and her unknown state jointly and sends the information through a classical channel to Bob. Although Alice's original unknown state is collapsed in the process of measuring the state of the entangled photon (due to the quantum non-cloning phenomenon), Bob can construct an accurate replica of Alice's state by applying a unitary operator."

Farbod Khoshnoud, I. I. Esat, S. Javaherian, B. Bahr, Edited by C.W. de Silva, Quantum Entanglement and Cryptography for Automation and Control of Dynamic Systems, Special issue of the Instrumentation Journal, Edited by C.W. de Silva, Vol. 6, No. 4, pp. 109-127, 2019.

"Quantum entanglement is realized by the Spontaneous Parametric Down-conversion process. Two entangled autonomous systems exhibit correlated behavior without any classical communication in between them due to the quantum entanglement phenomenon. Specifically, the behavior of a system, Bob, at a distance, is correlated with a corresponding system, Alice. In an automation scenario, the "Bob Robot" is entangled with the "Alice Robot" in performing autonomous tasks without any classical connection between them. Quantum cryptography is a capability that allows guaranteed security. Such capabilities can be implemented in control of

autonomous mechanical systems where, for instance, an “Alice Autonomous System” can control a “Bob Autonomous System” for applications of automation and robotics.”

Farbod Khoshnoud, Ibrahim I. Esat (Brunel), Marco B. Quadrelli (JPL), Dario Robinson (UPD Police), Quantum Cooperative Robotics and Autonomy, Special issue of the Instrumentation Journal, Edited by C.W. de Silva, Vol. 6, No. 3, pp. 93-111, 2019.

“Adopt the experimental aspects of quantum entanglement and quantum cryptography, and integrate these established quantum capabilities into distributed robotic platforms, to explore the possibility of achieving increased autonomy for the control of multi-agent robotic systems engaged in cooperative tasks. Experimental quantum capabilities are realized by producing single photons (using spontaneous parametric down-conversion process), polarization of photons, detecting vertical and horizontal polarizations, and single photon detecting/counting. Specifically, such quantum aspects are implemented on network of classical agents, i.e.,

classical aerial and ground robots/unmanned systems. With respect to classical systems for robotic applications, leveraging quantum technology is expected to lead to guaranteed security, very fast control and communication, and unparalleled quantum capabilities such as entanglement and quantum superposition that will enable novel applications.”

Farbod Khoshnoud, II Esat, CW De Silva, Marco B. Quadrelli (JPL), Quantum Network of Cooperative Unmanned Autonomous Systems, Unmanned Systems, Vol. 07, No. 02, pp. 137-145 (2019).

"Sending photons of light through nonlinear crystals produces correlated photon pairs, by splitting each photon into two half particles with each particle having the same level of energy, which results in entangled pairs. This entanglement is represented by photons, having both either horizontal or vertical polarization. This paper investigates collaborative robotic tasks of unmanned systems in a network where the agents are entangled. For instance, a leader robot sends two identical photons (e.g. with vertical polarization) to two follower robots/autonomous vehicles to communicate information about various tasks such as swarm, formation, trajectory tracking, path following and collaborative tasks. The potential advantages of quantum cooperation of robotic agents is the speed of the process, the ability to achieve security with immunity against cyberattacks, and fault tolerance, through entanglement. If a Quantum Network is implemented in a robotic application, it would present an effective solution…"

Farbod Khoshnoud, C. W. de Silva, and I. I. Esat, Quantum Entanglement of Autonomous Vehicles for Cyber-physical security, IEEE International Conference on Systems, Man, and Cybernetics, Banff, Canada, October 5–8, pp. 2655- 2660, 2017.

"This paper discusses collaboration of autonomous unmanned systems in the presence of cyber-physical attacks, inspired by Quantum Entanglement (QE). QE describes a quantum state of correlated pairs or groups of particles as a whole while interacting, even for the particles separated by a large distance. The QE is a physical phenomenon at the molecular level and below, and the effects of QE would quickly be lost at the macroscopic scale due to the interactions of the system with the environment and the exchange of thermal radiation. However, entanglement between

animals and in the biological domain have been suggested, such as the hypotheses of quantum computations, which rely on QE that is taking place in the brain, and quantum cooperation of two ants, or two distant butterflies. The present research discusses scenarios of autonomous systems maintaining their collaboration even if the communication is lost between them, and they are separated by a large distance…"

Hua-Ying Liu, Xiao-Hui Tian, Changsheng Gu, Pengfei Fan, et al., Drone-based all-weather entanglement distribution, National Science Review,Volume 7, Issue 5, Pages 921-928, 2020.

"The quantum satellite is a cornerstone towards practical free-space quantum network and overcomes the photon loss over large distance. However, challenges still exist including real-time all-location coverage and multi-node construction, which may be complemented by the diversity of modern drones. Here we demonstrate the first drone-based entanglement distribution at all-weather conditions over 200 meters (test field limited), and the Clauser-Horne-Shimony-Holt S-parameter exceeds 2.49, within 35 kg take-off weight. With symmetric transmitter and receiver beam apertures and single-mode-fiber-coupling technology, such progress is ready for future quantum network with multi-node expansion. This network can be further integrated in picture-drone sizes for plug-and-play local-area coverage, or loaded onto high-altitude drones for wide-area coverage, which adds flexibility while connecting to the existing satellites and ground fiber-based quantum network."

Djordjevic, Ivan. Quantum Communication, Quantum Networks, and Quantum Sensing / Ivan B. Djordjevic. London: Academic Press, 2022. Print.

“Quantum Communication, Quantum Networks, and Quantum Sensing represents a self-contained introduction to quantum communication, quantum error-correction, quantum networks, and quantum sensing. It starts with basic concepts from classical detection theory, information theory, and channel coding fundamentals before continuing with basic principles of quantum mechanics including state vectors, operators, density operators, measurements, and dynamics of a quantum system. It continues with fundamental principles of quantum information processing, basic quantum gates, no-cloning and theorem on indistinguishability of arbitrary quantum states. The book then focuses on quantum information theory, quantum detection and Gaussian quantum information theories, and quantum key distribution (QKD). The book then covers quantum error correction codes (QECCs) before introducing quantum networks. The book concludes with quantum sensing and quantum radars, quantum machine learning and fault-tolerant quantum error correction concepts. Integrates quantum information processing fundamentals, quantum communication, quantum error correction, quantum networks, QKD, quantum sensing, and quantum machine learning Provides in-depth exposition on the design of quantum error correction circuits, quantum communications systems, quantum networks, and quantum sensing systems Shows how to design the information processing circuits, stabilizer codes, CSS codes, entanglement-assisted quantum error correction codes Describes quantum machine learning”

Djordjevic, Ivan. Quantum Communication, Quantum Networks, and Quantum Sensing / Ivan B. Djordjevic. London: Academic Press, 2022. Print.

“In asymmetric algorithms , the encryption and decryption keys are different. Moreover, the decryption key cannot be determined from the encryption key, at least in any reasonable amount of time. For this reason, the encryption keys can even be made public because an eavesdropper will be unable to determine the decryption key. Public-key systems [5] are based on this concept. The encryption keys are made public in public-key systems, while the decryption key is known only to the intended user. The encryption key is then called the public key, while the decryption key is known as the secret (private) key.”

Cai, Wenqi. “Optimal Control Strategies for the Alignment Problem of Optical Communication Systems.” King Abdullah University of Science and Technology, 2020.

“Our proposed method for the mobile optical communication system has two steps: coarse alignment and fine alignment, as shown in Fig. 3.2. The position coordinates of the receiver in the image frame (x, y) is the input of the system. For a tracking error (pan error or tilt error) or a following error (angle error or distance error), the feedback PD control is applied to eliminate these errors. Then, to get rid of the effects of any disturbances, a fine alignment using the perturbation-based ESC is adopted to keep finding the optimal position, which allows getting the maximum received optical power Pmax at the receiver”

Cai, Wenqi. “Optimal Control Strategies for the Alignment Problem of Optical Communication Systems.” King Abdullah University of Science and Technology, 2020.

“When the receiver (color code) is recognized by Pixy2, a bounding box is drawn on it as shown in Fig. 3.3. This hard work of receiver detection and localization is handled by the image processing system inside the Pixy2 camera, and the position coordinates, block size, and color code of the detected receiver will be reported back to Arduino. In the Arduino, the information of position coordinates (x, y) is utilized to adjust the pan and tilt servos to keep the tracked receiver in the center of the image frame. The simple feedback PD control is used for both pan and tilt servos"

III. Problem Definition

With the capabilities that quantum cryptography presents, this could greatly benefit sectors such as the government/military that utilize unmanned aerial vehicles (UAV). These UAVs are susceptible to security attacks due to their storage of highly classified and important information. UAV-based healthcare systems can also benefit by implementing quantum communication as they too struggle with the confidentiality and privacy of medical records during data transferring [20]. Besides defense and healthcare, there are also the banking and utilities of each country that would greatly benefit from quantum cryptography. Banking and finance sectors control sensitive financial data daily, this makes them a desirable target for cyber attacks. The energy and utilities sector are also susceptible to attacks to critical infrastructure such as power grids, oil pipelines, and water treatment.

The main focus of the project was the manual placement and adjustment of the laser source and polarizers. It is essential that the laser or transmitter properly passes through the polarizers. Autonomously adjustable polarizers will be utilized to manipulate the position of the transmitted photons, to ensure a proper connection between the transmitter and the receiver with minimal human interaction. A UAV in combat must be able to quickly and abruptly change its

position in order to avoid oncoming attacks, and thus should provide a great deal of useful data collection in seeing if system integrity is maintained.

While the project shall be conducted at a small scale with close proximity with each transmitter and receiver, there was an issue regarding long distances and how it can affect the system integrity. Even within the lab, basic testing highlighted the issue that any slight variations in lighting would affect the visual tracking capabilities of the Microsoft LifeCam and will cause it to track multiple objects it believes to be the color hue chosen. This also ties in to the signal degradation regarding quantum communication due to turbulence and other factors that affect the signal-to-noise ratio [6][15]. To overcome the visual tracking issues by using a raspberry pi compatible camera alongside a QR Tag on the other bot for tracking as the tag will make it easier to track regardless of lighting conditions in the environment. Should the two system feedback loop be established, the laser required for the quantum communication will then turn on and allow the system to establish a link and verifying through a data logging program that tracks polarizations being received [12].

As many of the components needed to construct the project are big ticket items, the intended market being military/defense, many of the cost and technological constraints can be alleviated right away. Small, fragile components such as the beamsplitter cube can cost upward of $200 alongside other required components for quantum cryptography that can lead to a high cost. However, other components such as servos, cameras, and power supplies are less costly. While this can somewhat alleviate some of the cost constraints, there are also the constraints of technology. Most of the components used in the project, while state of the art, are relatively primitive. As it stands, most of the information gathered from these experiments are with hopes that future versions of many of the components will get smaller and inexpensive if they are to reach a broader market.

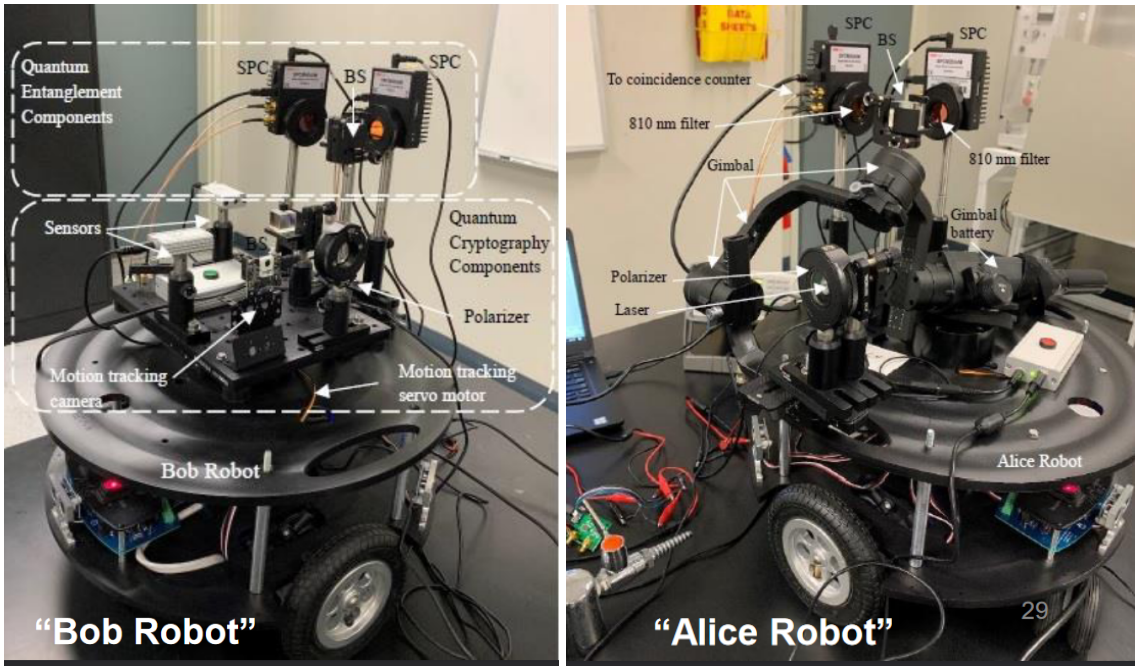


Fig. 3 Alice and Bob. [8]

IV. Materials and Methods

For the restoration of the IHerb robot, we used an arduino to run 2 brushless motors with 2 BLDC (brushless direct current) motor drivers, and a pixy2cam for motion sensing. We will also be using a PhantomX XL430 for the laser coarse alignment and a raspberry pi to run it. The robot itself was donated from a bankrupt company courtesy of Dr. Boskovich. We sourced a lot of our code foundation from ChaptGPT and had to make adjustments to it for our purposes. Our main methodology was trial and error, we were given very little info about the IHerb robot and so completed our own research as well as had assistance from Dr. Herder, Dr. Khoshnoud, and Anan, our department's technician.

V. Project Plan

With the general background of previous work done regarding quantum cryptography briefly explained, the focus will now switch to what the overall goals are for the current experiment. The general timeline for the project shall be carried on within the duration of the following senior project classes. Based on the work that has already been conducted in this experiment, a reconstruction of the systems of both the Bob and Alice robot will have to be done.

As a result of having multiple dynamic bodies (e.g. Alice and Bob ground vehicle), line of communication between these vehicles is critical as their position can change at any moment based on the environment and situation. To maintain effective communication between all bodies, the usage of fine and coarse tracking will be implemented to adjust the direction the quantum cryptographic components need to send each other messages. Pictured below is an example of one of the previous versions of the system to provide a visual aid on how the pan & tilt motors are used.

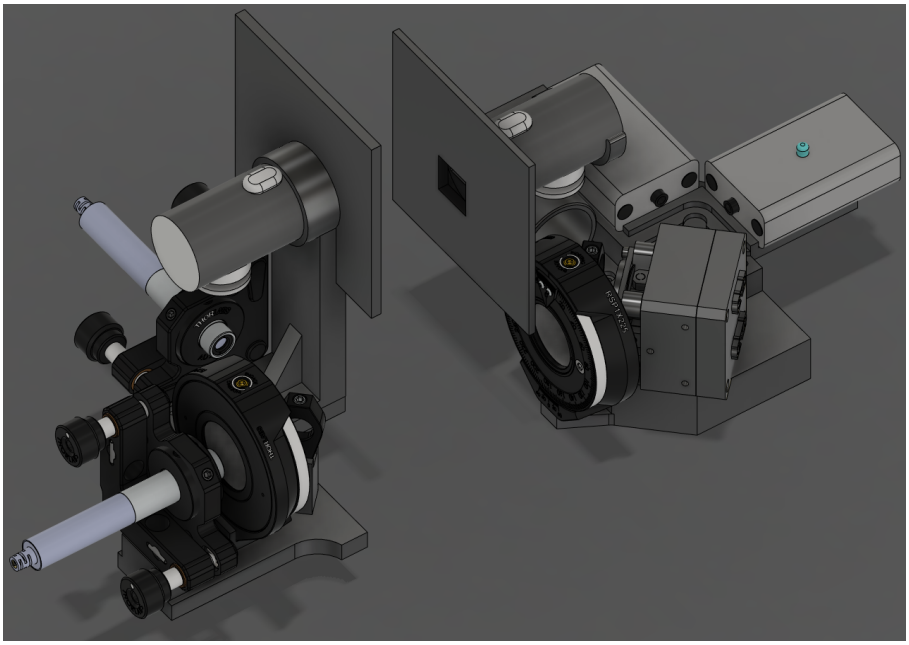


Fig. 4 Component View. [21]

Using the exploded view above, the hardware to be mounted on top of the pan and tilt turret going in order from 1 to 10, the components are as follows: Camera, Red Laser, Dichroic Mirror, QR Tag, Green Laser, Half Wave Plate, Single Photon Counter, Fast Steering Mirror, Position Sensing Detector, Beam Splitter. To achieve the design solution outlined above, the following approach will be taken:

1. Selection of Commercially Available Pan & Tilt Mechanism: A pan & tilt mechanism that meets the weight and torque requirements will be chosen to serve as the foundation of the design. This mechanism will provide the necessary range of motion for the alignment and tracking systems.

2. Integration of Coarse Alignment System: A camera will be mounted on the pan & tilt mechanism for coarse alignment purposes. The camera will utilize AR codes for tracking, allowing it to locate and maintain visual contact with the receiver. The feedback loop will be established to guide the pan & tilt mechanism based on the camera's input, ensuring accurate positioning.

3. Incorporation of Fine Alignment System: A fast steering mirror will be integrated into the design to facilitate fine alignment. The position sensor detector will provide feedback on the laser's position, allowing for precise adjustment to maintain optimal alignment with the receiver. This system will ensure accurate targeting and alignment during movement.

4. Implementation of Polarization Control: Half wave plates will be utilized to control the polarization of the laser source. By adjusting the orientation of the half wave plates, the polarization of the laser can be modified, enabling efficient and reliable communication between the transmitter and receiver.

5. Integration with Autonomous Robot: The pan & tilt mechanism, along with the alignment and communication components, will be mounted on an autonomous robot platform. This integration will allow the entire system to move autonomously while maintaining optical alignment and communication with the receiver. The weight distribution and balance of the components on the robot will be carefully considered to ensure stability and performance.

6. Verification and Testing: The design will undergo thorough verification and testing to ensure its functionality and effectiveness. Various scenarios and conditions will be simulated to evaluate the system's performance, including tracking accuracy, alignment stability, and communication reliability. Adjustments and refinements will be made as necessary based on the test results.

All the quantum portions of the robot that must be aligned to one another shall then be mounted directly on top of the Trossen Robotics PhantomX XL430 Pan & Tilt turret. With a payload of 500g and stall torque of 1.5 N.m, careful planning on what the components placed on the servo must take place to better make use of the available force. While the turret comes prebuilt, it is planned to be dismantled to strip it down to its bare essentials to reduce and stay within the weight limit. Towards the back of the turret, the Dynamixel U2D2 communication converter and its power hub board are to be mounted directly to the top of the robot and remove the shell of the turret alongside its aluminum breadboard that is installed on the top of the motors. For Alice, she will carry the laser diode and a half wave polarizing rotator plate while Bob will carry a half wave polarization rotator plate, beam splitter cube, and two sensors [14][8]. These components will be salvaged from the ThorLabs demonstration kit. Whether or not exactly if the weight of some of the components are within the weight limit of the servo’s payload, from basic observations it’s assumed that all these components on top of the servos will be too much for it to handle. For this instance a solution is to find some way to use another pair of servos and connect them either

alongside the XL430 to continue using it and the Lifecam camera or using just another pair of servos entirely.

The specific microcontroller in mind will be the Raspberry Pi 4b 4GB model as per the recommendation of Trossen Robotics in order to run the software needed to control their servos as well as receiving the signal decrypted by the quantum components to execute the digital command sent by the polarized photons [8]. In order to control the servos, a system of both coarse and fine alignment using feedback photodiode (PD) control and extremum seeking (ESC) control for each respectively is to be used [9] [13]. By having both tracking systems, a reduction of the QKD signal degradation by maintaining constant link between the quantum components [13].

The pan and tilt required a bit more involvement as there are multiple pieces of software needed to get the turret up and running. Firstly Ubuntu 20.04 must be installed into either a laptop or single board computer like the Raspberry Pi 4. This alone can take a bit of computer skills, especially for those who have little to no experience using a Linux-based operating system. For those using a laptop, it's assumed that it is more than likely a windows machine that must be properly set up for dual-booting to have the linux portion installed [17]. After this the specific install instructions of ROS Noetic for Ubuntu are then followed to create the environment for packages used by ROS to control the Turret [16][18]. When it came time to actually download the operating system into the RPi4 there were a few hiccups encountered and resolved. To give a bit of background, all the instructions given by Trossen Robotics to set up the RPi4 to work with the turret were followed, but almost immediately issues arose. Initially Ubuntu Server 20.04.5 LTS with Ubuntu Mate as the desktop environment was used as per the instructions, however the operating system in general was extremely sluggish opening a simple terminal or moving the window around the screen. This coupled with the software from Trossen not running, it was decided to move to another desktop environment. We downloaded an older version of PiOS

documented by the ROS wiki to work with Noetic, hopes were high but the software by Trossen was not compatible with the OS. After plenty of research finding the lightest weight desktop environment for the Server image of Ubuntu, it was decided that LXDE as the desktop environment was the superior choice due to its speed and low resource usage that would alleviate some of the resource constraints. Thankfully, when redownloading all the software, no issues arose, thus there was a successful connection allowing the camera to work flawlessly with the pan & tilt and immediately got to work on the next portion of the project.

Using a CMOS pixel sensor, more specifically a Microsoft Lifecam connected via USB cable to the Raspberry Pi 4, coarse object tracking between Alice and Bob ground vehicles will be achieved as a QR Tag will be mounted on both Alice and Bob as a sort of anchor for the CMOS to focus on [5]. The QR tag was specifically chosen due to issues regarding the color tracking software of the XL430 running on the RPi4. As of this time, it is unclear if there weren't enough resources available for the program to run properly or other hardware constraints, but it is suspected that maybe the integrated graphics of the RPi4 isn’t strong enough to run that portion of the program.

As of right now, the fine alignment components are working on a test bench but have yet to implement them with the coarse alignment components. Below is a photo taken in the lab of how the components were mounted. One of the issues observed with this setup was that the photon source (laser) is fairly underpowered and does not work when it is past 300 mm from the closed loop system. This can easily be resolved with a stronger laser, however there is then the issue of a potential eye hazard as it will then be too powerful. More research is currently being conducted to see if there is a safer alternative in resolving this issue.



Fig. 5 Prototype for fine tracking system

Using the schematic of the Airborne Entangled-Photon Source (AEPS) pictured below a reconstruction of system shown where a polarization entangled photon source which will then communicate directly to Bob and Alice will first be done on a test bench to create a rough estimation on where each component will be placed on the robots themselves [5]. To accurately measure each entangled photon, a special program that measures coincidence rate over time to further refine the signals being sent and received between all systems. As progress continues with the program and research, updates in the testing methodology to explain what was done and what actions were taken to achieve them will be noted.

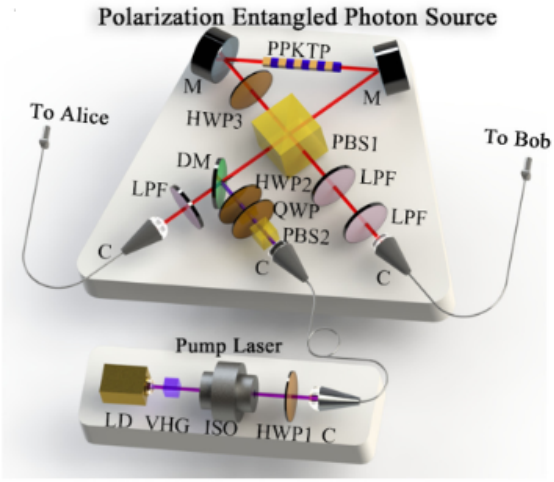


Fig. 2 AEPS Diagram. [9]

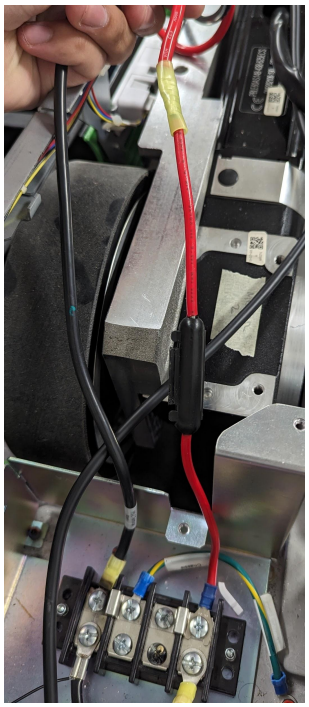
iHerb Restoration

To utilize the motors and chassis of the iHerb robot, several components must be acquired, as most other parts will be obtained from the robot itself. The essential components to purchase beforehand include: an Ardunio (this could be a Uno or Mega as we used) or similar microcontroller, two DC brushless motor drivers that can support at least 48 Volts and 15 Amps for each motor, Dupont wire for connecting from the Arduino to the motor driver, a Ferrule crimp and connector kit to connect the motors to the motor driver and power delivery, at least a 15 Amp inline fuse of at least 14 gauge wire, some eyelet connectors and crimps, Pixy2 Camera, and a computer with the Arduino IDE for the code.

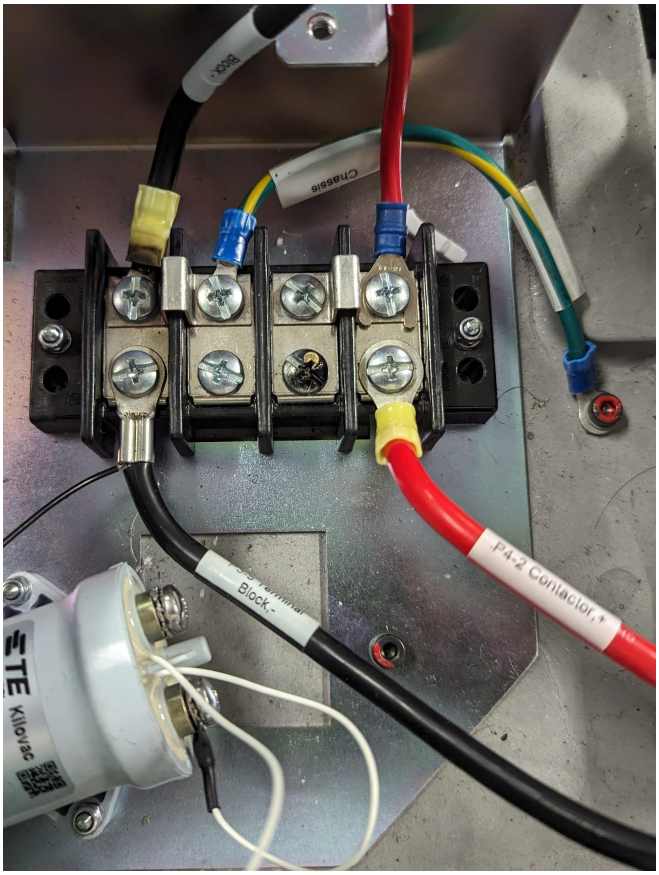
The initial step involves removing the screws from the upper portion of the robot's shell to access the internal components. It is strongly recommended to take photographs of the robot's original state before proceeding with the disassembly, to have reference images if needed. If there is a battery inside, turn off the circuit breaker switch then unplug and remove the battery by removing the screws on the side that hold it in place. Afterward, carefully remove all components

from the robot without damaging any cables or parts. Unplug all components connected to the PCB in the front before moving on with the two motors for the wheels as it will make it easier to reach some bolts that would be hidden. Remove all levers used by the lift in the center of the robot as none will be needed alongside the lift motor that is located towards the front middle of the robot by removing the bolts that hold them all down. To remove the PCB, there are a few bolts towards the bottom of the mounting plate that makes it easy to lift it out. The PLC located next to the PCB is mounted via velcro straps, make sure all connections are not damaged or removed from it. Underneath the PLC, there is a terminal block that we will use later alongside some cables that will be salvaged. There is a high current relay near the terminal block that could also be removed as it will not be used in this configuration

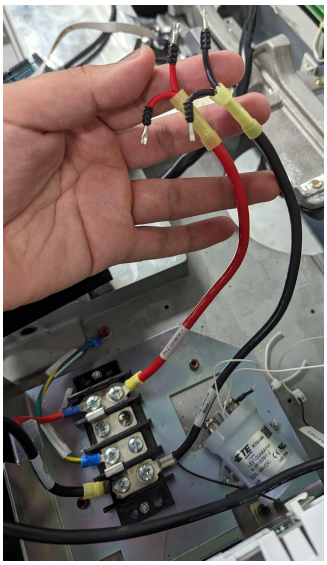
At this stage, the robot's interior should be completely empty, except for a few cables from the circuit breaker and some connected to the terminal block. There will be a short cable connecting the high current relay to the terminal block, a cable connecting the other connection on the high current relay to four smaller wire gauges before linking back to the negative side of the terminal block, a cable connecting to the circuit breaker and battery through the powerpole connector, and a green cable grounded by the robot's chassis. We will only utilize the cable for the circuit breaker and the cable connected to the high current relay with four smaller wire gauges; everything else can be removed except the green ground cable. Once the terminal block is entirely empty (excluding the green ground cable), we can proceed to the next step. Cut the end of the positive cable (colored red) that is connected to the circuit breaker and use a crimp connector to connect the inline fuse together.



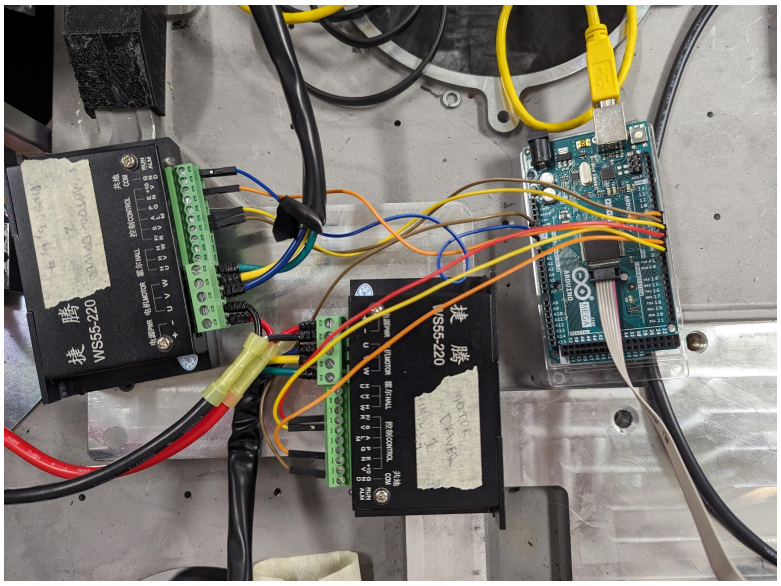
Strip the other end of the inline fuse cable that is free and crimp an eyelet connector to it, then connect it back to one end of the terminal block. This configuration ensures that the positive cable from the circuit switch is protected by a fuse before reaching the terminal block, preventing potential hazards from overcurrent. Connect the negative cable from the circuit breaker switch (colored black) to the opposite end of the terminal block. Next, use the cable with the four smaller wires and connect its positive end adjacent to the positive cable from the circuit breaker on the terminal block and its negative end adjacent to the negative cable from the circuit breaker on the terminal block.

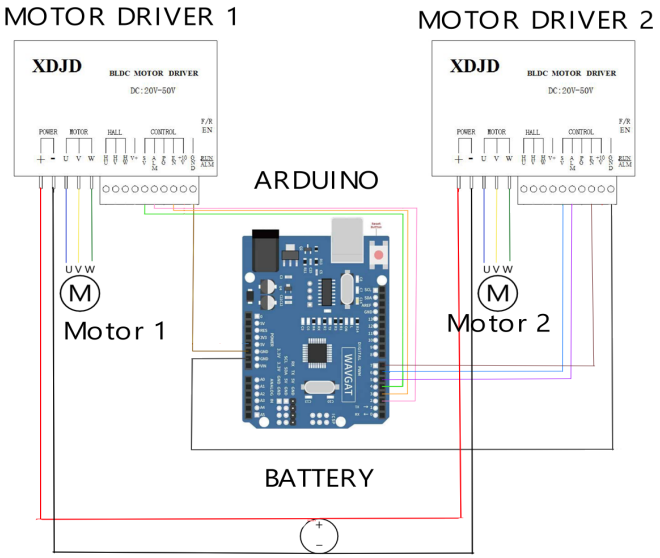


Now, while this cable has 4 smaller wires, it terminates to a connector that would normally connect to the PCB for power control, however we will not be using this connector anymore for this configuration. Cut the end connector off from the four wires, leaving enough slack for the four wires as we will be using them to power the motor driver later. Making sure the correct wire gauge is chosen and that there is plenty of slack on the wires, crimp the Ferrule connectors to each end of the four wires correctly making sure it is a firm connection before we continue. At this point, you will have two positive cables with Ferrule connectors that are crimped to a thicker cable connected to the terminal block next to the positive cable from the circuit breaker. There will also be two negative cables with Ferrule connectors that are crimped to a thicker cable connected to the terminal block next to the negative cable from the circuit breaker.

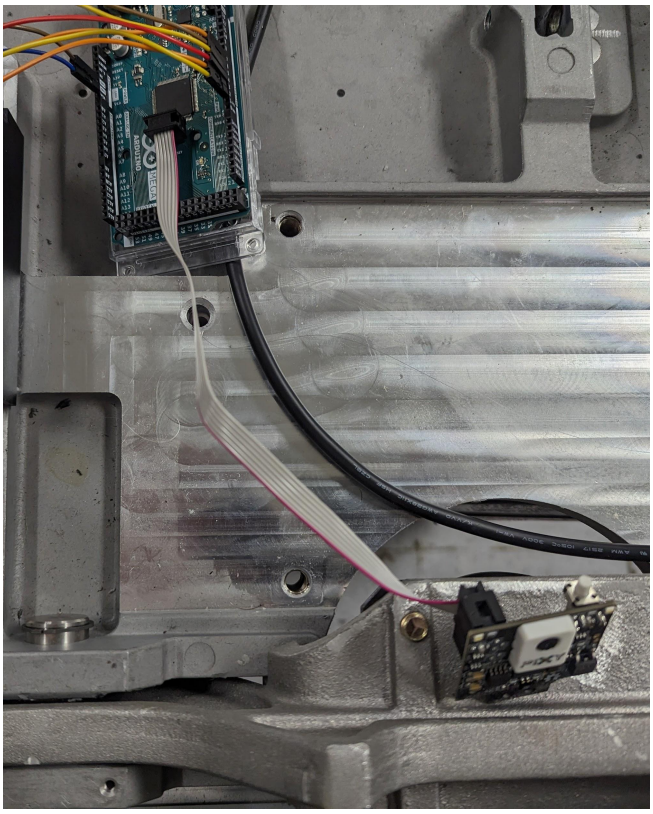


Moving on to the next step, reattach and connect the motors to the motor drivers. After creating the necessary cables, remount the two wheel motors back onto the chassis to enable movement. Exercise great care when bolting them back on to avoid damaging or stripping the bolts. Each motor will have three cables: Blue (U), Yellow (V), and Green (W) on the motor driver. Connect these cables to a connector originally used on the PCB, which will no longer be used. Cut each end with enough slack, then crimp Ferrule connectors on each end to simplify connection to the motor driver. Position the motor drivers where the PCB was initially mounted, making it easier for the power cables to reach and connect to them. A positive and negative cable must be connected to each motor driver. Refer to the simple wiring diagram ab to see where each cable from the motor driver to the arduino is connected.





One important aspect not covered in the wiring diagram is the Pixy2 camera. To connect the Pixy2 camera to the Arduino Uno, use the included ribbon cable from the Pixy2 box and connect it to the ICSP connector located at the bottom of the Uno.



After this, if each cable is connected exactly as shown in the wiring diagram, copy and paste the code below to the Arduino IDE. At this point, there are a few more housekeeping things to do before completion. We must first set a signature or color that the Pixy2 cam will track to move the robot. First, download PixyMon V2 fro

VI. SCHEDULE

The overall plan for this and next semester is to work on implementing the new polarizers that will automatically adjust itself depending on where the laser source is and fix the issue about the camera detecting multiple sources that have the same color hue as the laser. There is also the issue of fixing the issue of the detector losing proximity of the laser when it is too far away.

Table 1: Gantt Chart			
	June	July	
Week 1	iHerb Breakdown	Started adding course alignment and fine alignment imagery parts	
Week 2	Motordrivers installed Motor drivers controlled via potentiometer	Rapid prototyping of Quantum imagery mounting brackets via 3D Printing	
Week 3	Motordrivers connected to Arduino and set up to run with Arduino code	Quantum imagery brackets mounted course alignment setup Fine alignment setup	
Week 4	Setup Pixy2Cam tracking with Arduino program	Poster and PowerPoint slides created to	

With the summer break coming soon, it is the hope of this group to allocate plenty of uninterrupted time to completely focus on working on the project by finishing everything regarding the fine and coarse alignment so that progress on the ground vehicle to have it up and running for the coming semester is completed.

VII. Conclusions

The future impact of quantum communication and cryptography is undeniable. Providing the absolute best security by preventing any malicious interference and eavesdropping, there are still many obstacles that need to be addressed. While most experiments have been conducted by creating a point-to-point link, issues regarding beam diffraction from free-space tests and entry cost to build the necessary infrastructure for more large-scale tests will pose an issue for quick

adoption of the technology [11]. Regardless, quantum cryptography has the potential to revolutionize secure communication and has a wide range of applications across various sectors. Despite the challenges associated with implementing quantum cryptography in robotics and autonomous systems, the advancements in technology promise a future where quantum communication and cryptography become integral components of secure communication systems.

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